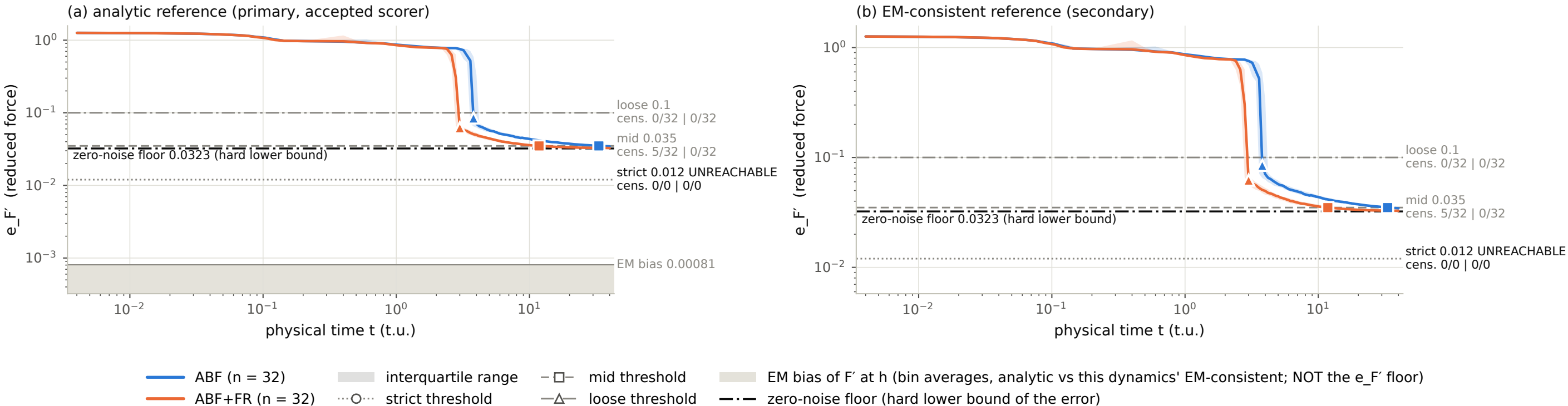


Gateway family matched_free_energy/alpha1 · N = 2048 · T_N = 40 t.u.

n_steps = 1,600,000 · h = 2.5e-05 · B = N n_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Mean-force error $e_{F'}$ vs physical time



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Markers: first save after which the seed-MEDIAN curve stays at or below the threshold at every later save (persistent crossing of the median curve; the per-seed tau statistics are in summary.json). Censored thresholds have no marker; 'cens. a | b' next to each threshold = seeds whose own tau is censored (never persistently met by u = 1), ABF | ABF+FR.

Errors on the eval window [-1.5, 1.5] (151 grid nodes; $e_{F'}$ function space on the Gauss-Legendre sub-grid); Gamma = M / (C + 1).

$e_{F'}$ has a deterministic zero-noise floor 0.03227 (the within-bin variation of F'_{ref} , steep at the gate): $e_{F'^2} = \text{RMS}(\text{Gamma} - \text{bin-averaged } F'_{ref})^2 + \text{floor}^2$ for EVERY estimate, so a threshold below the floor is unreachable by construction (strict); 'both censored' there is not a tie. The shaded band is the EM bias of F' (0.00081), not the floor.

eqb_plot_config/2 · config alpha1.json · 2026-10-10 14:20 UTC